Exam in Signals and Transforms (1TE746)

January 13, 2025

Instructions

General:

- **If you have passed either Part A (Questions 1–3) or B (Question 4–6) previously during the current academic year, you do not have to answer the corresponding part in this exam.**
- Use a separate page for each question and number the pages;
- Write your LADOK anonymization code on each page;
- **Clearly motivate your answers, missing motivations will give deductions;**
- Always simplify your answer as much as possible;
- Clearly mark your final answers (e.g., using double underlines or a frame);
- Always provide the correct unit, missing units will give deductions;
- Make sure that your calculator is set to radians.

Allowed Aids:

- The course's formula sheet;
- Template for Bode diagrams;
- One (1) single-sided A4 page with your own, hand-written notes on Chapters 1–5;
- One (1) single-sided A4 page with your own, hand-written notes on Chapters 6–9;
- Calculator;
- **No old exams, exercises, lecture notes, or other aids not listed above!**

Grading:

Parts A (Questions 1–3) and B (Questions 4–6) are graded individually according to the following grading criteria. To pass each part, you need to either

- obtain at least 2 points in the sub-questions a) and b) in all three questions and 8 points overall (all questions) or
- obtain at least 2 points in the sub-questions a) and b) in two of three questions and 12 points overall (all questions).

The final grade for each part, rounded to one decimal, is based on to the total points from all questions and calculated according to

$$\text{Grade} = \min\left(2.5 + 2\frac{\text{Points} - 8}{8}, 5\right).$$

The final grade is the mean of the two partial grades, rounded to the next whole number.

Part A: Continuous-time

Question 1: Transform methods

Figure 1 shows two periodic signals $x(t)$ and $y(t)$ as well as two sets of exponential Fourier series coefficients c_n and d_n.

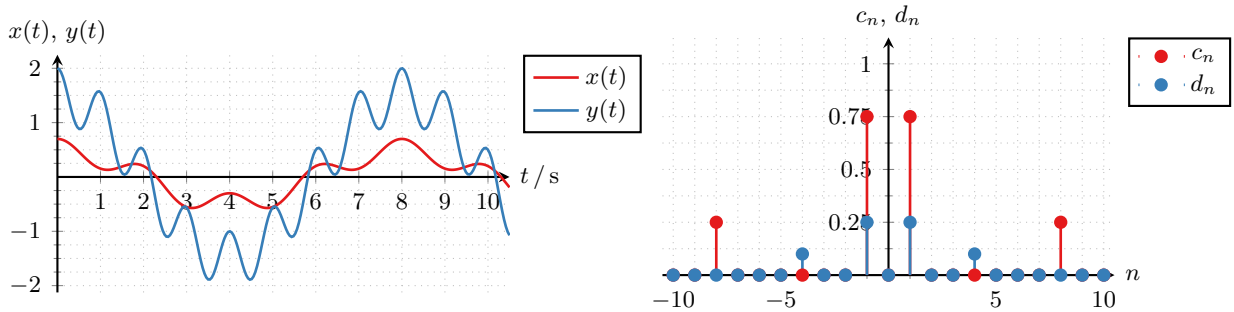

Figure 1: Signals $x(t)$ and $y(t)$ as well as Fourier series coefficients c_n and d_n.

a) (2p) Determine which Fourier coefficients correspond to which signal. Motivate your answer.
b) (2p) Determine the Fourier transform for the *new* signal

$$z(t) = \sum_{n=-\infty}^{\infty} d_n e^{\mathrm{j}\, n20t}.$$

with d_n as in Figure 1. Sketch $Z(\omega)$ for the interval $-100\,\mathrm{rad/s} \leq \omega \leq 100\,\mathrm{rad/s}$.
c) (2p,†) Determine the Laplace transform for the signal

$$v(t) = y(t)\, u(t).$$

Question 2: LTI systems

Consider the stable system $H(s)$ with Bode diagram in Figure 2 below.

a) (2p) Determine the output signal $y(t)$ if the input to the system is

$$x(t) = 2\left(e^{\mathrm{j}(0.8t+2\pi/3)} + e^{-\mathrm{j}(0.8t+2\pi/3)}\right).$$

Simplify as much as possible.
b) (2p) Determine the system's transfer function $H(s)$.
c) (4p,†) Determine the output signal $y(t)$ of the system for the input signal

$$x(t) = e^{-2t}\cos(3t)u(t).$$

Question 3: Filtering theory

Consider the brick-wall specifications for a lowpass filter shown in Figure 3.

a) (2p) Determine the minimum order for a Chebyshev type II filter fulfilling these requirements.
b) (2p) Give three desirable properties of real filters and discuss why it is difficult to achieve these at the same time.
c) (2p,†) Given the signal

$$z(t) = x(t) + y(t),$$

is it possible $x(t)$ and $y(t)$ from $z(t)$ using *real* linear, frequency-selective filters? Motivate your answer.

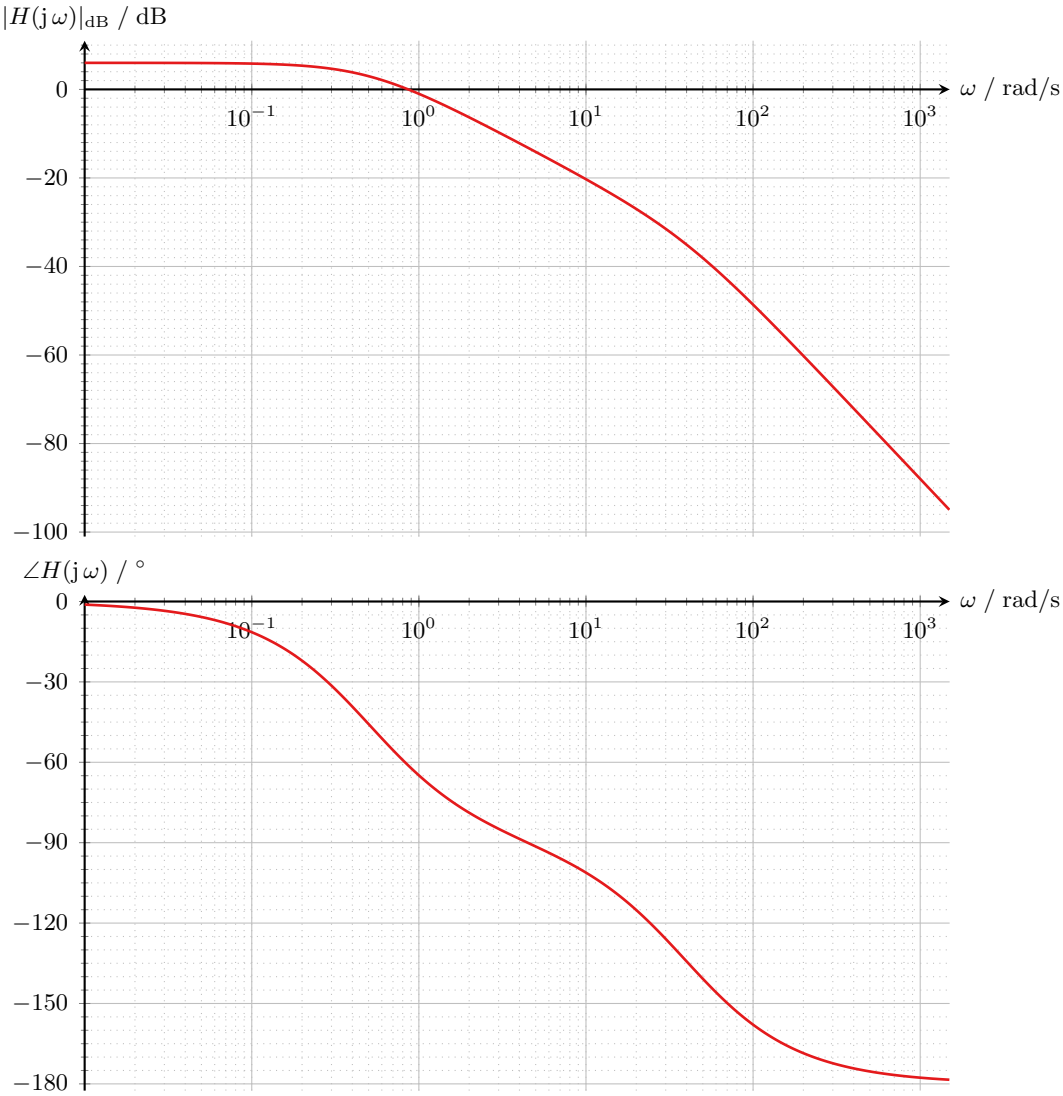

Figure 2: Bode diagram.

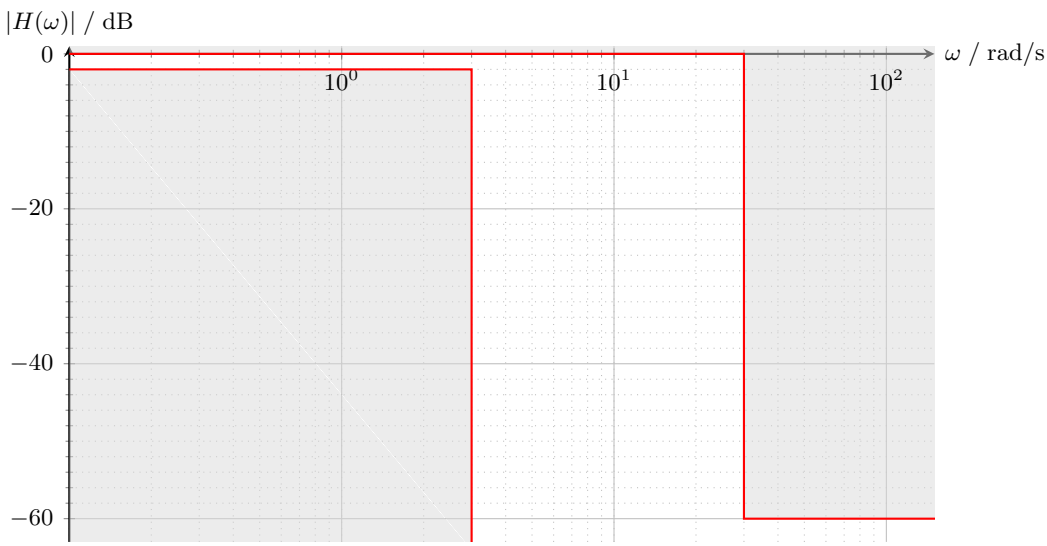

Figure 3: Brick-wall specifications.

Part B: Discrete-time

Question 4: Transform methods

The samples $x[0], x[1], \ldots, x[K-1]$ with $K = 20$ of a signal $x[k]$ were used to calculate the signal's discrete Fourier transform $X[l]$. This gave the result shown in Figure 4. Assume that great care was exercised when choosing the number of samples K. No particular window was used.

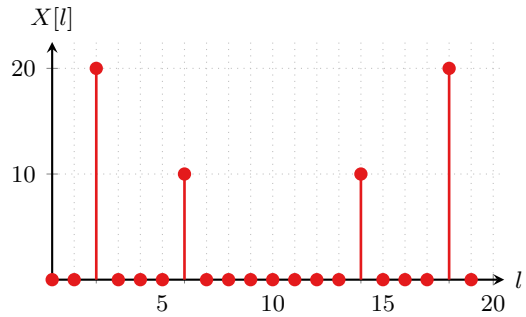

Figure 4: Discrete Fourier transform of the signal $x[k]$.

a) (2p) Determine the signal $x[k]$. Simplify as much as possible.
b) (2p) The discrete Fourier transform suffers from two effects called *smearing* and *leakage*. Explain these terms.
c) (2p,†) Assume that $x[k]$ was obtained by sampling a continuous-time signal $x(t)$ using $\omega_s = 400\,\text{rad/s}$. Sketch the spectrum $X(\omega)$ for the interval $-200\,\text{rad/s} \leq \omega \leq 200\,\text{rad/s}$.

Question 5: LTI systems

Figure 5 shows the pole-zero map for a discrete-time, linear, time-invariant system $H(z)$. Furthermore, it is known that $H(e^{\mathrm{j}\,0}) = 1$.

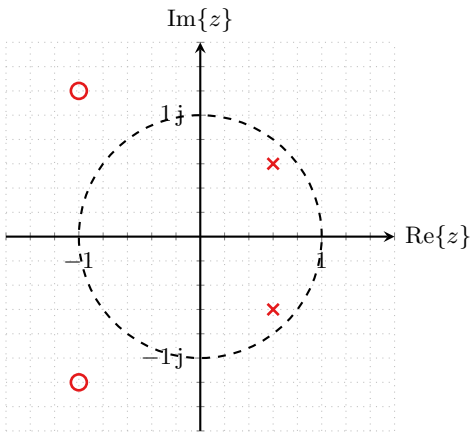

Figure 5: Pole-zero map for the system $H(z)$.

a) (2p) Is the system bounded-input, bounded-output stable? Motivate your answer.
b) (2p) Determine the system's transfer function $H(z)$ in terms of z^{-n} (negative form). Simplify and expand as much as possible.
c) (4,†) Determine the system's impulse response $h[k]$.

Question 6: Sampling and reconstruction

The spectrum for the continuous-time signal $x(t)$ is shown in Figure 6.

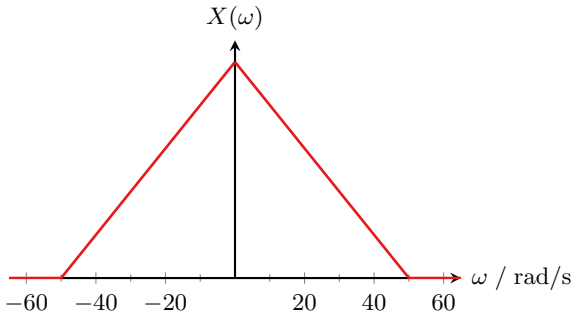

Figure 6: Spectrum of the signal $x(t)$.

a) (2p) Determine the lower bounds of the sampling frequency ω_s that allow for perfect reconstruction of the two signals

$$y(t) = x(2t),$$
$$z(t) = 2x(t).$$

b) (2p) Let $\omega_s = 80\,\text{rad/s}$. Sketch the spectrum of the sampled signal

$$x_s(t) = x(t)\,s(t)$$

for the interval $0\,\text{rad/s} \leq \omega \leq 160\,\text{rad/s}$. Recall that

$$s(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_s)$$

is the sampling signal (impulse train).

c) (2p,†) Explain the term *quantization*. How does it affect reconstruction of a continuous-time signal from its samples?